

Raymond Chen

Senior Software Engineer

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Professional Experience

Twilio

January 2020 - February 2025

Lead Software Engineer

- Led the development of a **HIPAA-compliant telemedicine platform** with **AI-powered triage**, reducing patient intake time by 40%.
- Implemented **code splitting** and **lazy loading** for secondary features to ensure quick initial loading of core features with using **Next.js**, **React**, and **TypeScript**.
- Architected a microservices-based backend with **RabbitMQ/Kafka** for event-driven architecture, **Redis** for caching, and **sharded databases** (MongoDB, PostgreSQL) for scalability and low-latency performance.
- Implemented **event-driven architecture** for real-time features to ensure responsiveness and separation of concerns with using **Java/Spring Boot** and **Node.js/Express**.
- Deployed services using **Docker** and **Kubernetes**, with global **load balancing** and centralized **monitoring/logging** (Prometheus, Grafana, ELK Stack).
- Implemented secure **authentication** (JWT, OAuth2.0), **role-based access control** (RBAC), and **E2EE** protocols to ensure system security.
- Architected and deployed scalable backend microservices using **AWS**, leveraging **EC2** for computing, **S3** for secure video storage, **Lambda** for serverless functions, and **API Gateway** for efficient API management.
- Automated **CI/CD pipelines** with **GitHub Actions** and **Jenkins**, enabling efficient testing and deployment workflows.

Roofstock

April 2017 - December 2019

Senior Software Engineer

- Led the migration to a headless **CMS**, achieving a 40% reduction in content update times and facilitating faster scalability for future expansions.
- Collaborated with data scientists to implement customer segmentation models, resulting in a 12% reduction in customer churn through personalized experiences.
- Conducted **A/B testing** on e-commerce workflows, identifying key pain points and driving a 15% increase in conversion rates.
- Automated membership billing processes and optimized **CI/CD pipelines**, significantly enhancing deployment frequency and system reliability.
- Performed high-load testing of the platform to evaluate scalability with concurrent users, ensuring stability during peak usage periods.
- **Monitored** and **optimized** real-time latency for video, audio, and chat, maintaining acceptable performance levels under diverse network conditions.

Software Engineer

- Developed **inventory management tools** that allowed businesses to track stock levels, manage waitlists for out-of-stock items, and monitor sales analytics.
 - Implemented the user-friendly interfaces for tracking inventory and managing waitlists with using **React.js** and **TypeScript**.
 - Employed **Redis** as a caching layer for faster access to frequently queried inventory and sales data.
 - Managed and queried transactional data to provide accurate insights into stock levels and sales analytics using **MySQL**.
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Skills

Backend: Java/Spring Boot, TypeScript, Hibernate, Maven, Gradle, Kotlin, Saas, J2EE, Servlets, Node.js/Express, Nest.js, Python/FastAPI, PHP/Laravel

Frontend: HTML5, CSS3, JavaScript/TypeScript, Bootstrap, Tailwind CSS, React.js, Angular, Vue.js, Next.js, Svelte/SvelteKit

Database: MySQL, PostgreSQL, MongoDB, DynamoDB

Cloud/DevOps: Docker, Kubernetes, AWS, Azure, CI/CD (Jenkins, CircleCI, etc), Terraform, GitHub Actions

APIs: RESTful APIs, GraphQL, gRPC

Architecture: Microservices Architecture, Micro-frontend, Scalable Web Applications

Testing/QA: Unit Testing (Jest, Mocha), Integration Testing, E2E Testing (Playwright)

Tools/Methods: Git, Agile/Scrum, Kanban, Unit Testing (Mocha, JUnit), E2E Testing

Education

University of Houston

Bachelor's Degree in Computer Science

Baltimore, MD

July 2011 - December 2015
